

Problem regarding variation quantum eigensolver

<https://github.com/Qiskit/qiskit-tutorials/issues/515>

ROW 45

Problem regarding variation quantum eigensolver #515



Closed



OChiB opened on Feb 7, 2019 - edited by OChiB

Edits ...

Informations

- Qiskit version: 0.6
- Python version: 3.7
- Operating system: Microsoft

I want to program my own variation quantum eigensolver for my own practice.

The Hamiltonian considering is a two level system, so Pauli-z gate.

The ansatz state is rotated by the ry gate.

The problem is, the program works fine, if I use the defined function once, which returns the measurement result for a certain parameter angle. For example, if I set the angle to pi, it returns -1.0 as expected.

But, it fails, if I use this function iteratively, for example, by making use of for-loop.

In this case, the measurement result converges to 0, but not to -1.

What I think is, the classical register does not initialize the measurement result, if I use it repeatedly.

How can I fix it?

Thank you very much in advance!

This is my code:

```
#Testing variational quantum eigensolver

from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
from qiskit import available_backends, execute, register, get_backend
from qiskit.tools.visualization import circuit_drawer
from qiskit.tools.qi.qi import state_fidelity
from qiskit import Aer

backend = Aer.get_backend('qasm_simulator')

#additional packages
import matplotlib.pyplot as plt
import numpy as np
from scipy.optimize import minimize
import math

#define Quantum register: 1 qubit
qreg = QuantumRegister(1)
creg = ClassicalRegister(1)
#define quantum circuit for 1 qubit
qc = QuantumCircuit(qreg, creg)

#define expectation value of H
def expec(theta):
    #define number of samples/measurements
    samples = 10000

    #ansatz state
    qc.ry(float(theta[0]),qreg[0])

    #measurement (sampling)
    qc.measure(qreg[0], creg[0])

    backend = Aer.get_backend('qasm_simulator')

    job = execute(qc,backend,shots=samples)
    result = job.result()

    #calculate expectation
    c0 = result.get_counts(qc).get('0',0)/samples
    c1= result.get_counts(qc).get('1',0)/samples

    print(c0 - c1)
    return c0 - c1
    #store result

#single test
'''
test = expec([np.pi])
print(test)
```

Assignees

chunfuchen

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



```

print(1000)
"""
#iterative test with plot
"""
params_range = np.linspace(0.0, 2 * np.pi, 25)
data = [expec([angle]) for angle in params_range]

plt.xlabel('Parameter value')
plt.ylabel('Expectation value')
plt.plot(params_range, data)
plt.show()
"""

#minimization algorithm using Nelder-Mead Simplex Method
"""
theta_init = [2.0]
minimum = minimize(expec, theta_init, method='Nelder-Mead', options= {'maxiter':10, 'initial_simplex': np.array,

print(minimum.x)
"""

```

 **OChiB** added **bug** on Feb 7, 2019

 **attp** assigned [chunfuchen](#) on Feb 7, 2019



chunfuchen on Feb 8, 2019 · edited by chunfuchen

Edits ▾ Contributor ...

[@OChiB](#) do you mind reformat your script? you can use `"""python"""` to wrapper your script.

I guess that there is something wrong in the calculation of your expectation, it should be $c_1 - c_0$.

On the other hand, Aqua provides APIs to build VQE. Here is the equvalent script as yours.
and I do get the result is `-1`.

```

from qiskit import Aer
from qiskit.quantum_info import Pauli
from qiskit_aqua import Operator, set_aqua_logging
from qiskit_aqua.algorithms import VQE
from qiskit_aqua.components.variational_forms import RY
from qiskit_aqua.components.optimizers import NELDER_MEAD
import logging

# set to INFO to see the evaluation progress
set_aqua_logging(logging.INFO)

# prepare qubit operator
H = Operator(paulis=[[1.0, Pauli.from_label('Z')]])

# get backend
backend = Aer.get_backend('qasm_simulator')

# setup VQE
var_form = RY(H.num_qubits, depth=1)
optimizer = NELDER_MEAD(maxfev=100)
algo = VQE(H, var_form, optimizer, 'paulis')

# run it
result = algo.run(backend, shots=8192)
print(result['eigvals'][0])

```

```
# run it
result = algo.run(backend, shots=8192)
print(result['eigvals'][0])
```



OChiB on Feb 8, 2019

Author ...

@chunfuchen Thank you for your reply!

Regarding to the expectation calculation. It shouldn't be wrong, since this expectation value return -1 for the angle pi. And I think this is not the main issue, since the essential problem lies on the iterative algorithm, not in the calculation of the expectation value itself (since otherwise the expectation value should give 1 or -1, but not 0).

I tried to run your code, but I get

TypeError: '>=' not supported between instances of 'QuantumRegister' and 'int'

I have updated the qiskit, qiskit-aqua, and qiskit-chemistry, but this also doesn't work.



chunfuchen on Feb 8, 2019 · edited by chunfuchen

Edits Contributor ...

@OChiB I am using qiskit 0.7.2, do you mind upgrading your qiskit to the newest version?

here is my env

```
qiskit                0.7.2
qiskit-aer             0.1.1
qiskit-aqua            0.4.1
qiskit-chemistry       0.4.2
qiskit-terra           0.7.0
```



chunfuchen on Feb 8, 2019

Contributor ...

@OChiB I found a bug in your code, you should move the declaration of `QuantumCircuit` into your `def expect(theta)` function otherwise, the `qc` is a reference, which means you keep appending the ry gate.

After I move the declaration of `QuantumCircuit` into your `def expect(theta)` function, it converges to -1



OChiB on Feb 9, 2019

Author ...

@chunfuchen Oh thank you very much !! I could fix it!!



OChiB closed this as [completed](#) on Feb 9, 2019